

METHODOLOGY REPORT FOR E-Commerce Conversion Prediction Challenge

1. DATA LOADING AND INSPECTION :

- Initial verification included shape, data auditing, checking for missing values and duplicates.

2. EDA :

- Histograms were made for all the numerical columns to see their distribution
- Box plots were made to detect outliers.
- Target(Converted column) variance was identified and imbalance was noticed and thus the use of `class_weight='balanced'` was done.

3. DATA CLEANING:

- Duplicates if any were removed (It was observed that the data had no duplicates).
- Missing values were imputed with median.
- The interquartile range method was used for dealing with outliers
- The same was done for `public_test` and `private _ test` to prevent data leakage.

4. FEATURE ENGINEERING:

- `Product_Page_Ratio` was calculated to measure browsing intent concentration.
- `Time_Per_Page` was calculated to measure user concentration and intention on page.
- `Purchase_Discount` was calculated to measure if customers who received discounts came back again or not?
- `Income_Age` was calculated to normalize earning power with stage of life. Which income bracket and age bracket spends more?

5. MODEL DEVELOPMENT:

- Preprocessing for done using `Standard Scalar` (for numeric data) and `OneHotEncoder` (for text data)
- Classification : `class_weight= ' balanced'` to compensate for the imbalance in dataset, `c=10` to control regularization and allow better fitting of model, `max_iter=5000` for convergence on entire data set

6. EVALUATION:

- F1 score was calculated for training and validation datasets.

- Train and Validation F1 score was compared side by side to access overfitting.
- Grid Search for threshold optimization was done.

The final prediction was done using `predict_proba` and thresholded at optimal value during validation.